

Learned Indexes From the One-dimensional to the Multi-dimensional Spaces: Challenges, Techniques, and Opportunities

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Abstract

Recently, the class of learned index structures has emerged as one form of database indexes that make use of Machine Learning (ML) techniques. The learned indexes designed for the one-dimensional space have demonstrated improvements in both the query processing time and the index size. Observing the advantages of one-dimensional learned indexes, various learned indexes have been proposed for the multi-dimensional space. This class of learned indexes is termed “Learned Multi-dimensional Indexes.” This tutorial on learned indexes is designed based on our long survey article on the subject [4]. In this tutorial, we use a taxonomy to categorize over 100 learned one- and multi-dimensional indexes with more focus on the class of learned multi-dimensional indexes. The goal of this tutorial is to explain the fundamental techniques behind state-of-the-art learned one- and multi-dimensional indexes with emphasis on the latter, and identify the ongoing challenges and future opportunities for research in this domain.

CCS Concepts

• Information systems → Data management systems.

Keywords

Machine Learning for Database Systems, Learned Index, Learned Multi-dimensional Index

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1 Introduction

Machine Learning (ML) techniques have been successfully applied to enhance the performance of core database systems components, e.g., indexes [59] and query optimizers [84]. Particularly, treating a database index structure as an ML model is a key observation in building *learned* index structures. Initially, learned indexes have been designed for the one-dimensional case [27, 35, 59]. Due to

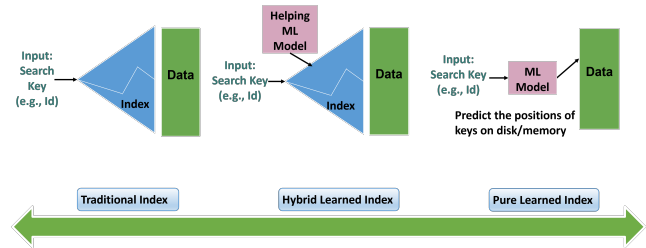


Figure 1: Spectrum of learned index structures.

the performance benefits of one-dimensional learned indexes, the concept has been extended for the multi-dimensional space [58]. Numerous learned multi-dimensional indexes have been proposed in the literature, e.g., [3, 4, 24, 28, 63, 66, 90, 96]. Learned index structures can be broadly categorized into pure and hybrid learned indexes. Pure learned indexes are primarily designed to replace traditional index structures, e.g., to replace the B-tree [9, 17], the R-tree [42], or the Bloom filter [13]. On the other hand, hybrid learned indexes combine ML models with traditional indexes to build an ML-enhanced index structure. The spectrum of learned indexes is given in Figure 1.

In this tutorial, we use a taxonomy that we have developed in our survey paper [4] (also highlighted in Figure 2) to classify over 100 learned one- and multi-dimensional indexes. The concept of learned indexes has been extended into the multi-dimensional space due to the promising performance of the learned one-dimensional indexes. The following four approaches have been adopted to design various learned multi-dimensional indexes:

- Approach 1: Traditional multi-dimensional indexes are augmented with ML models.
- Approach 2: Multi-dimensional data are projected into the one-dimensional space and a learned one-dimensional index is built in the projected space.
- Approach 3: A learned one-dimensional index is applied once per dimension, on one or more dimensions of the multi-dimensional space without using any projection functions.
- Approach 4: Learned multi-dimensional indexes are designed from scratch by leveraging a suitably arranged data layout.

The various learned one-dimensional indexes have influenced the design choices for learned multi-dimensional indexes. As a result, in this tutorial, an overview of learned indexes for the one-dimensional space will be covered first, and this will serve as the foundation for learned indexes for the multi-dimensional space.



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In the taxonomy, in the context of learning the index, we refer to the task of learning the mapping from keys to positions inside a data set as the learned index. First, we differentiate between **Immutable vs. Mutable learned indexes** based on the capability of supporting dynamic inserts/updates. We further classify the mutable learned indexes based on **Fixed vs. Dynamic data layouts**. If the data layout is fixed before the construction of an index, we refer to the index as having a fixed data layout. In contrast, in the context of dynamic data layouts, ML models help re-arrange the data layout during the index construction phase.

Next, the learned indexes are categorized based on the underlying data space: **One-dimensional vs. Multi-dimensional space**. Both the learned one- and multi-dimensional indexes are categorized to identify their position in the spectrum (Figure 1) of learned indexes: **Pure vs. Hybrid**. In the context of mutable pure learned indexes, we also distinguish the indexes based on their adopted strategies for supporting new data insertions: **In-place vs. Delta buffer insertion strategies**. The hybrid learned indexes are categorized based on their traditional index component (e.g., B-tree, R-tree, etc.). Finally, the taxonomy sub-categorizes further the class of learned multi-dimensional indexes based on how they deal with the underlying space into **Projected vs. Native space**. For example, some learned multi-dimensional indexes use Space Filling Curves (SFC, for short) [88, 104] to map the multi-dimensional space into the one-dimensional space, and then apply a learned model to learn the mapped data in the one-dimensional space.

The evolution of research related to learned indexes is presented in Figure 3. It is evident from Figure 3 that the field of learned one- and multi-dimensional indexes is evolving rapidly. As a result, this tutorial will cover the current state of research in this highly active field with more focus on learned indexes for the multi-dimensional case, and present several open challenges for future research.

2 Tutorial Outline

We plan to deliver a 1.5-hour lecture-style tutorial. Moreover, we split the tutorial into the following parts:

- Introduction and Historical Background (10 Minutes)
 - Learning the Index vs. Indexing the Learned Models
 - Evolution of Learned Indexes
- Part 1: Learned Indexes for the One-dimensional Space (20 minutes)
 - Immutable vs. Mutable Indexes (5 minutes)
 - Fixed vs. Dynamic Data Layouts (5 minutes)
 - Pure vs. Hybrid Indexes (5 minutes)
 - In-place vs. Delta Buffer Insertion Strategies (5 minutes)
- Part 2: Learned Indexes for the Multi-dimensional Space (55 minutes)
 - Motivation and Challenges (5 minutes)
 - * Projected vs. Native Space
 - Learned Multi-dimensional Indexes (50 minutes)
 - * Immutable Indexes (10 minutes)
 - Immutable Pure Indexes (5 minutes)
 - Immutable Hybrid Indexes (5 minutes)
 - * Mutable Indexes with Fixed Data Layouts (20 minutes)
 - Pure Indexes with In-place vs. Delta Buffer Insertion Strategies (10 minutes)

- Hybrid Indexes (10 minutes)
- * Mutable Indexes with Dynamic Data Layouts (20 minutes)
 - Pure Indexes with In-place vs. Delta Buffer Insertion Strategies (10 minutes)
 - Hybrid Indexes (10 minutes)
- Part 3: Open Challenges for Future Research (5 minutes)

3 Historical Background

3.1 Learning the Index vs. Indexing the Learned Models

In this tutorial, we first differentiate between the concepts of **Indexing the Learned Models vs. Learning the Index**. In the context of indexing the learned models, e.g., [7], the focus is to index the learned (i.e., trained) ML models so that we can distinguish among these models in an efficient way during query processing. On the other hand, in the context of learning the index, we refer to the task of learning the mapping (i.e., learned index) from keys to positions inside a data set. We will give some historical background about several early approaches, e.g., [52, 107] for indexing the learned models to contrast them against the more recent trend of learning the index.

3.2 Evolution of Learned Indexes

We will present and discuss the evolution of research for both one- and multi-dimensional learned indexes using Figure 3. The Figure 3 is adapted from our survey paper on learned indexes [4]. Here, we have grouped the papers on learned one- and multi-dimensional indexes based on their publication years. Moreover, we have used the $\rightarrow$ symbol to indicate if a later paper is related to an earlier one. The $\leftrightarrow$ is used when there is a line crossing. Additionally, a $\circ$ symbol is used to denote a copy of an earlier paper in a later year. One- and multi-dimensional indexes are differentiated using the $\square$ and $\triangle$ symbols, respectively.

4 Part 1: Learned Indexes for the One-dimensional Space

The Recursive Model Index (RMI, for short) [59] is considered the first instance of a “Learned Index” for the one-dimensional space. In [59], the key insight is that traditional indexes (e.g., the B-tree) can be perceived as an ML (i.e., predictive) models. For example, given an input search key, a B-tree predicts (i.e., searches) the key’s position in a logical sorted array. Hence, an ML model can potentially serve as a replacement for a traditional index. RMI proposes the idea of learning the Cumulative Distribution Function (CDF) to learn the key-to-position mapping. However, learning this mapping is challenging, and might not be accurate. As a result, an error correction mechanism is used after each ML model prediction. Notice that a key assumption is that the data is totally ordered (i.e., sorted) in the one-dimensional space. After the introduction of RMI, various indexes have been proposed in the literature addressing different issues related to the learned indexes: supporting dynamic inserts/updates, e.g., [27, 35, 129], supporting concurrency, e.g., [116]. Moreover, in the context of hybrid learned indexes, ML-enhanced

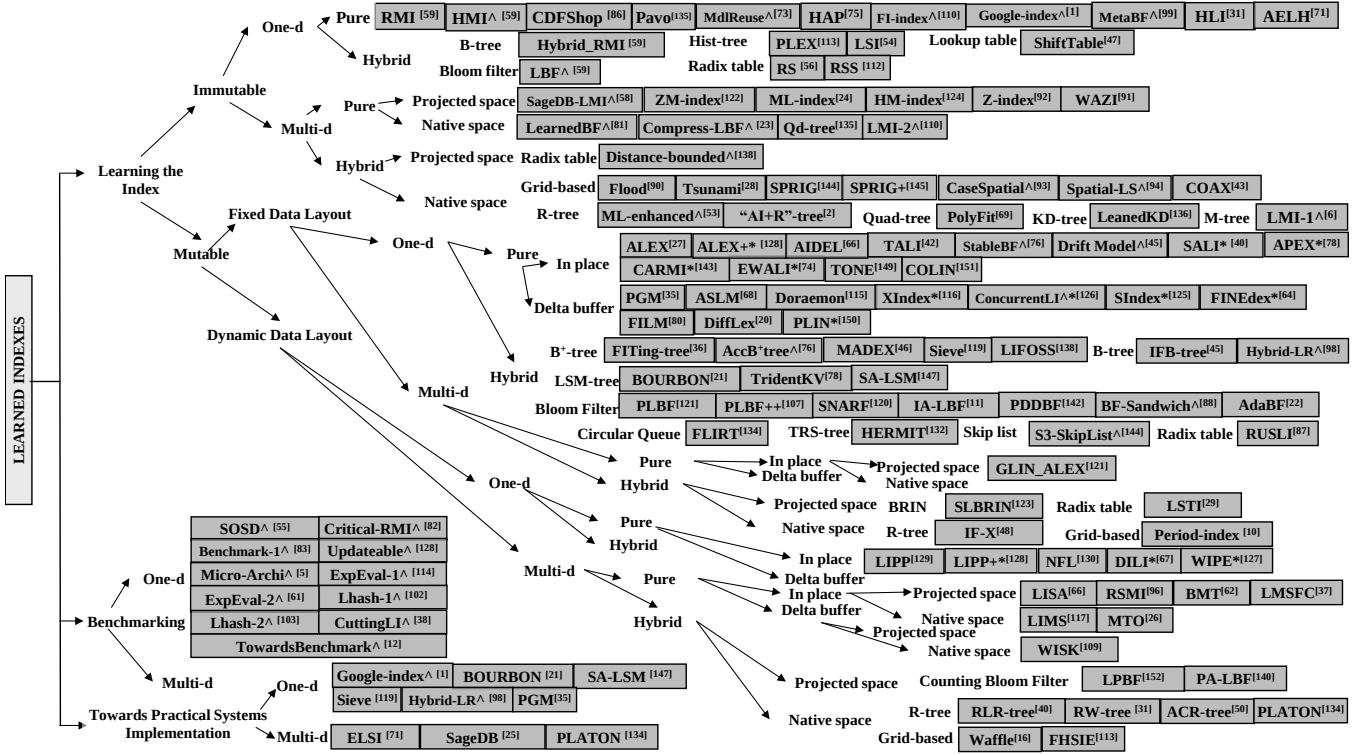


Figure 2: Taxonomy of learned indexes. A wedge ($\wedge$) symbol indicates that we have assigned a name to refer a particular paper or an index whenever a fixed name is not given by the original authors of that paper. An asterisk (*) symbol is used if the index natively supports concurrency. The hybrid learned indexes are categorized based on their underlying traditional data structures. The end of a branch indicates that there are no papers in that category as of the time this article has been written.

variants of traditional index structures, e.g., Bloom filter [13], LSM-tree [79], Skiplist [95]) have been introduced [21, 87, 143].

In Part 1, we will introduce the taxonomy, and present the fundamental concepts of learned index structures. Moreover, we will cover the existing learned indexes designed for the one-dimensional space. Particularly, we will follow the taxonomy, and explain the core concepts of a representative learned one-dimensional index from each class of indexes. Part 1 will serve as the foundation for Part 2 of the tutorial.

4.1 Immutable vs. Mutable Indexes

In the taxonomy (Figure 2), in the context of learning the index, we will first differentiate between immutable and mutable learned indexes based on the capability of supporting dynamic inserts/updates. The class of *Immutable* one-dimensional indexes contains 18 indexes [1, 19, 30, 47, 54, 56, 59, 70, 72, 74, 85, 98, 108, 111, 112, 132]. On the other hand, in the taxonomy, the class of *Mutable* one-dimensional indexes contains 48 indexes [11, 20–22, 27, 35, 36, 39, 41, 44–46, 64, 65, 67, 68, 73, 75–78, 80, 86, 87, 97, 106, 115, 116, 118–120, 125–128, 128–131, 133, 137, 141–143, 146, 148–150]. We will present and discuss the core ideas of representative immutable and mutable one-dimensional learned indexes.

4.2 Fixed vs. Dynamic Data Layout

The class of mutable learned indexes are further classified into two categories based on the data layout during the index construction phase. We will present the core ideas of representative mutable one-dimensional learned indexes (e.g., BOURBON [21], LIPP [129]) to explain the concepts of fixed and dynamic data layouts.

4.3 Pure vs. Hybrid Indexes

In the taxonomy, both immutable and mutable learned indexes are categorized into two types: Pure vs. Hybrid (see Figure 1). The class of immutable learned one-dimensional indexes contains 11 pure and 7 hybrid indexes. For example, we will cover RMI [59], and Hybrid_RMI [59] as the representative indexes to illustrate the pure and hybrid categories, respectively.

4.4 In-place vs. Delta Buffer Insertion Strategy

The class of mutable pure indexes are further categorized into two types based on the strategy for supporting new data insertions. For example, we will present the ALEX [27], and the PGM [35] as the representative mutable pure indexes to explain the concepts of In-place and Delta Buffer insertion strategies, respectively.

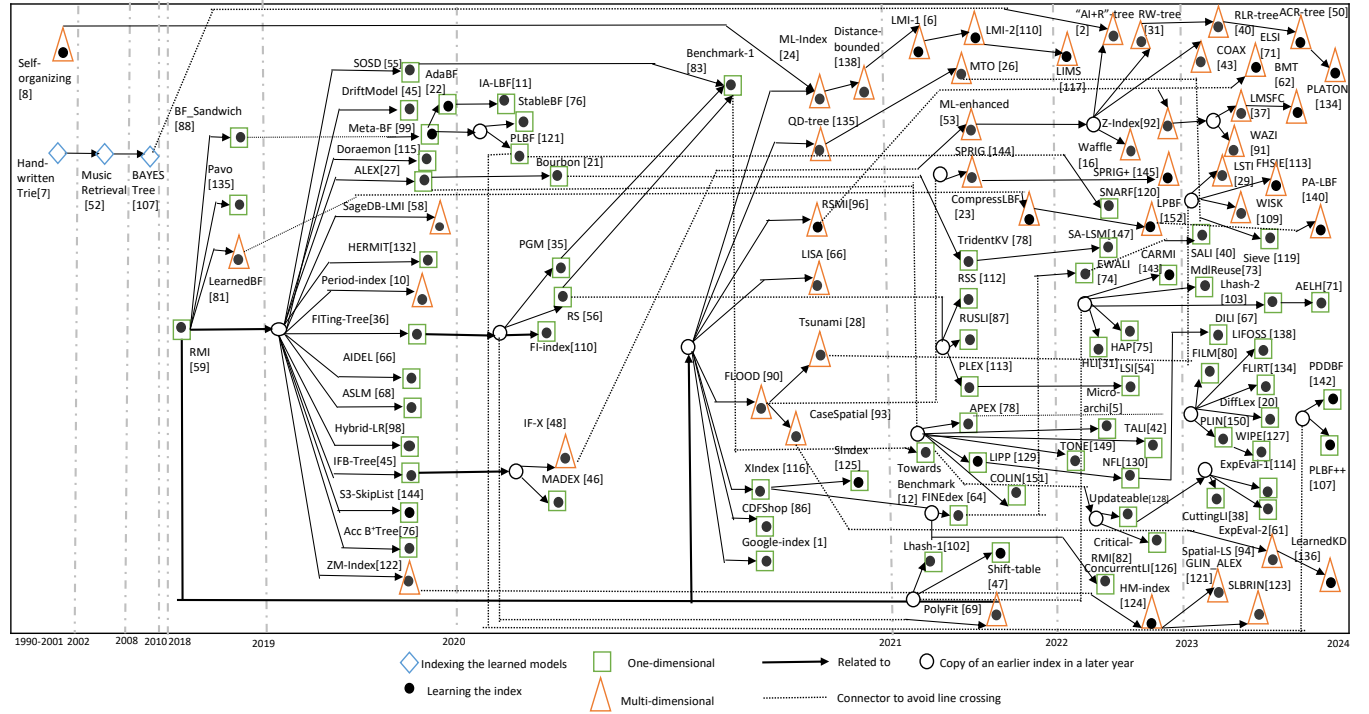


Figure 3: Evolution of learned indexes. Lines connecting between the various learned indexes reflect dependence of these indexes on earlier work.

5 Part 2: Learned indexes for the Multi-dimensional space

In Part 2, we will cover the state-of-the-art by presenting the motivation and challenges, and explaining the core concepts of the learned multi-dimensional indexes. Moreover, we will use the taxonomy to group similar indexes together, and highlight the similarities and differences within each group. Notice that we will also highlight the open branches in the taxonomy during our presentation.

5.1 Motivation and Challenges

Although the term learned multi-dimensional indexes has been introduced recently, one of the earliest papers on a distribution-aware index structure for spatial (i.e., multi-dimensional) data can be found in [8]. We will begin Part 2 by presenting the motivation behind extending the concept of learned one-dimensional indexes into the multi-dimensional space. We will discuss the additional challenges related to the development of learned multi-dimensional indexes. Moreover, we will also introduce the taxonomy categories that are only applicable to the learned multi-dimensional indexes: Projected vs. Native Space.

5.2 Immutable Pure Indexes

The class of *Immutable Pure* learned indexes, e.g., [23, 24, 58, 81, 91, 92, 110, 122, 124, 135], operate on the projected space by using a Space-filling Curve order (e.g., Z-order [89], Hilbert order [49]).

Some indexes in this class, e.g., the ZM-index [122] and Qd-tree [135], operate on the original representation of the data (i.e., native space). We will explain the core concepts behind this class of learned indexes.

5.3 Immutable Hybrid Indexes

The class of *Immutable Hybrid* learned indexes, e.g., [2, 6, 28, 43, 53, 69, 90, 93, 94, 136, 138, 144, 145], is classified based on the underlying traditional index component used (e.g., R-tree [42], Quad-tree [105]). We will present representative indexes from each group (e.g., grid-based, tree-based) of hybrid indexes.

5.4 Mutable Indexes with Fixed Data Layout

In the taxonomy, there are several indexes [10, 29, 48, 121, 123] in this category of mutable indexes with fixed data layout. Moreover, the pure indexes with fixed data layout are further categorized into two branches based on their new data insertion strategy: In-place vs. Delta Buffer. On the other hand, the hybrid indexes are further categorized based on their underlying traditional index structures. We will present the core ideas and the similarities/differences of the learned indexes in each of the branches.

5.5 Mutable Indexes with Dynamic Data Layout

Indexes in this category [16, 26, 31, 37, 40, 50, 62, 66, 96, 109, 113, 117, 134, 140, 152] are classified in the category of mutable indexes with dynamic data layout. Moreover, the pure indexes with dynamic

data layout are further categorized into two branches based on their new data insertion strategy: In-place vs. Delta Buffer. On the other hand, the hybrid indexes are further categorized based on their underlying traditional index structures. We will cover the representative learned multi-dimensional indexes from each of the branches according to the taxonomy.

5.6 A Summary of ML Techniques, Supported Query Types, and Practical Systems Integration

We will present a summary of the various ML techniques used for learned one- and multi-dimensional indexes. Moreover, a summary of the supported query types (e.g., point, range, kNN, and join queries) for each of the 40+ learned multi-dimensional indexes will be included. On the other hand, several learned one-dimensional indexes have been successfully integrated into practical systems [1, 21, 35, 97, 118, 146]. We will also discuss similar efforts in the context of learned multi-dimensional indexes [25, 71, 134].

6 Part 3: Open Challenges and Future Directions

In Part 3, we will discuss several open challenges and future research opportunities related to learned one- and multi-dimensional indexes. The open challenges and directions for future research are as follows:

6.1 The Lack of Total Ordering and Error Bound for the Multi-dimensional Case

Due to the lack of total ordering in the multi-dimensional space, it is challenging to provide an error bound in case of ML model mispredictions. As a result, a class of learned multi-dimensional indexes projects multi-dimensional data into the one-dimensional space to impose an ordering. However, some learned multi-dimensional indexes might select one of the dimensions to impose an order in native space. Although various methods have been proposed to address this challenge, there is still room for improvement.

6.2 Choice of ML models

In the context of learned indexes, to reduce the impact of ML model training time, storage overhead, and prediction latency, simple ML models are adopted whenever possible. However, choosing an ML model for a particular learned index design is a challenging task. As a result, a learned index should avoid using complex ML models so that the model building time and prediction latency do not become bottlenecks for achieving low index construction time and high query processing performance, respectively.

6.3 ML Model Re-training

In the context of learned indexes, changes in the underlying input data/query distribution should be detected as soon as possible, and a model re-training process should be triggered when necessary. For example, exploring the concept of Machine Unlearning [60] in the context of mutable learned indexes is an interesting research direction

6.4 Supporting Dynamic Inserts/Updates

Although mutable learned one- and multi-dimensional indexes can support inserts/updates, further research is needed to investigate the advantages and disadvantages of each approach, e.g., as in [14].

6.5 Supporting Concurrency

Only a few proposed methods (marked with an * symbol in the taxonomy Figure 2) discuss the issue of concurrency in the context of learned indexes. As a result, future research in this area should treat the issue of concurrency as a first-class citizen while designing learned one- and multi-dimensional indexes.

6.6 Index Compression

Both one- and multi-dimensional indexes have demonstrated significant benefits in terms of reduced storage requirements. Although the advantages of a learned multi-dimensional bloom filter for index compression are studied in CompressLBF [23], exploring the potential benefits of index compression using learned one- and multi-dimensional indexes in the context of other index types is an interesting direction for future research.

6.7 Security

The issue of security in the context of learned indexes is little explored. The impact of poisoning attacks has been discussed in a recent study [57]. Particularly, indexes, e.g., PGM [35], that are designed with a worst-case guarantee are expected to perform well in the presence of adversarial queries. Exploring this direction in the context of one- and multi-dimensional learned indexes is an open research topic.

6.8 Benchmarking for the Multi-dimensional Case

There are extensive benchmarking studies for various learned one-dimensional indexes [5, 12, 38, 55, 61, 82, 83, 102, 103, 114, 128]. However, similar effort is still missing in the context of learned multi-dimensional indexes.

6.9 Theoretical Analysis for the Multi-dimensional Case

There are a few theoretical studies [15, 32–34, 139] that mathematically analyze the reasons behind the performance gain of learned one-dimensional indexes over traditional indexes. Theoretical analysis of the various components of learned multi-dimensional indexes is needed to better understand the benefits and limitations of existing techniques.

6.10 Leveraging Modern Hardware for Learned Indexes

The benefit of natively implementing a learned one-dimensional index on a GPU has been presented in [151]. Similar investigation in the context of other learned one- and multi-dimensional indexes is an interesting direction for future research.

7 Related Tutorials

There is a prior tutorial on the subject of learned multi-dimensional indexes [3] that has been offered by the same authors of this tutorial.¹ However, the prior tutorial [3] covers methods only till the end of the year 2020. Hence, this tutorial serves as a significant extension over that earlier tutorial [3] by using a more sophisticated taxonomy to cover the newer varieties of learned multi-dimensional indexes. A survey paper on learned indexes by the same authors of this tutorial can be found in [4]. Another related tutorial can be found in [51]. The tutorial [51] presents the initial results related to learned indexes. However, in [51], only the Flood [90] index structure was presented as the representative learned multi-dimensional index. In a recent tutorial on spatial query optimization [147], seven learned spatial (i.e., multi-dimensional) indexes have been included in the context of I/O cost estimation. Another recent tutorial on machine learning for databases [18] covers only the ML-enhanced variants of R-tree index. In contrast, in addition to covering learned one-dimensional indexes as background, this tutorial will provide a comprehensive overview of learned multi-dimensional indexes by including over 40 learned multi-dimensional indexes. A series of related tutorials on big spatial data can be found in [99–101]. However, the focus of these series of tutorials is not on learned index structures.

8 Target Audiences and Goals

Intended Audience. This tutorial is intended for a broad category of students, academics, researchers, and practitioners with basic knowledge of data structures, algorithms, and machine learning. The tutorial is designed to be self-contained in providing all the necessary background. We only assume basic understanding of fundamental data indexing structures (e.g., the B-tree, the R-Tree, and the Bloom Filter).

Learning Goals. The target outcomes include:

- Understanding the fundamental concepts of learned index structures for the one- and multi-dimensional spaces.
- Familiarity of state-of-the-art learned one-dimensional indexes and thorough coverage of the state-of-the-art of learned multi-dimensional indexes using a taxonomy.
- Knowing the evolution of learned indexes using a timeline figure and recognizing their relationships and differences.
- Identifying the open challenges and future research opportunities in the area of learned one- and multi-dimensional indexes.

9 Short Biographies

Abdullah Al-Mamun is a PhD candidate at the Department of Computer Science, Purdue University. His research interests are in the area of Machine Learning for Database Systems, particularly, in the field of Learned Multi- and High-dimensional Index Structures.

Jianguo Wang is a tenure-track assistant professor at the Department of Computer Science, Purdue University. He received his Ph.D. degree in Computer Science from UC San Diego. His research interests include disaggregated databases and vector databases. His research has won multiple prestigious awards, including the ACM

SIGMOD Research Highlight Award, the NSF CAREER Award, and the IEEE TCDE Rising Star Award.

Walid G. Aref is a professor at the Department of Computer Science, Purdue University. His research interests are in extending the functionality of database systems in support of emerging applications, e.g., spatial, spatio-temporal and graph databases. He is also interested in query processing, indexing, and data streaming. He has served as the Editor-in-Chief of the ACM Transactions of Spatial Algorithms and Systems (ACM TSAS), an editor of the VLDB Journal and the ACM Transactions of Database Systems (ACM TODS). Walid has won several best paper awards including the 2016 VLDB ten-year best paper award. He is a Fellow of the IEEE, and a member of the ACM.

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